



LECTURE # 2 – PROBABILITY



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WHAT IS PROBABILITY?

People talk loosely about **probability** all the time:

- “What are the chances the Pakistan will win against India this weekend?”
- “What’s the chance of rain in the match?”

For scientific purposes, we need to be more specific in terms of defining and using probabilities



Rolling a 14



Heads



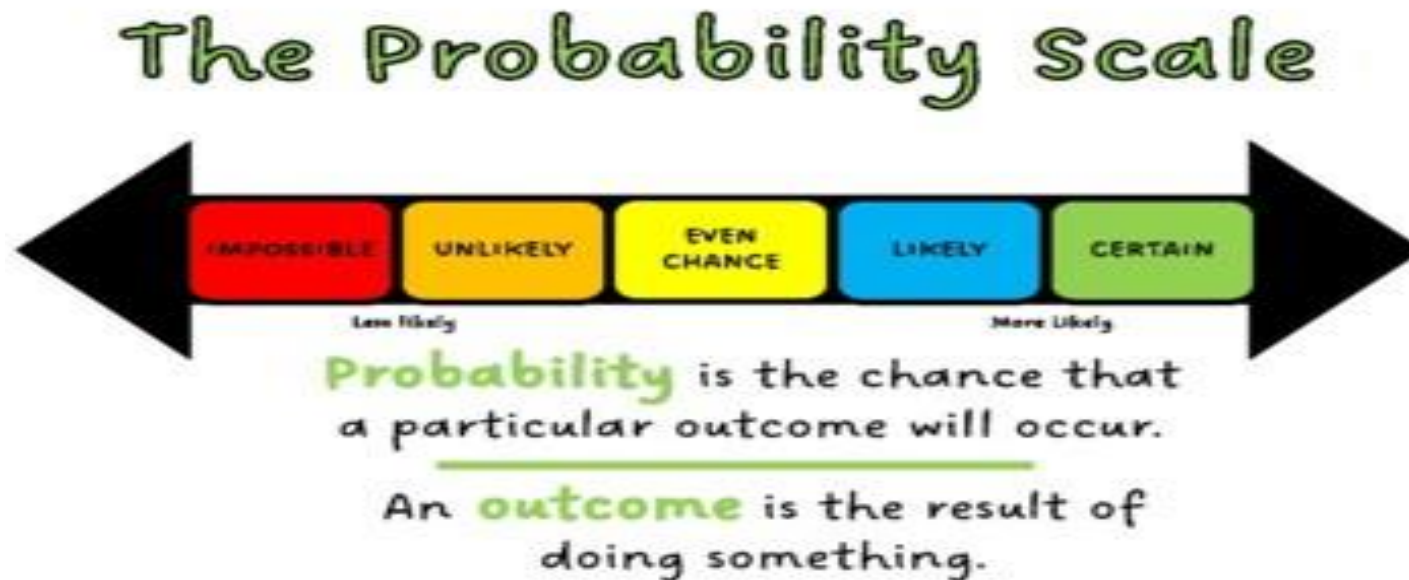
The sun will rise



...HOW LIKELY SOMETHING IS...

Probability is how likely something is to happen. You might also hear it called **chance**.

Probability can be expressed as fractions ($\frac{2}{3}$), decimals (0.35) or percentages (45%) ,
or on a probability scale.

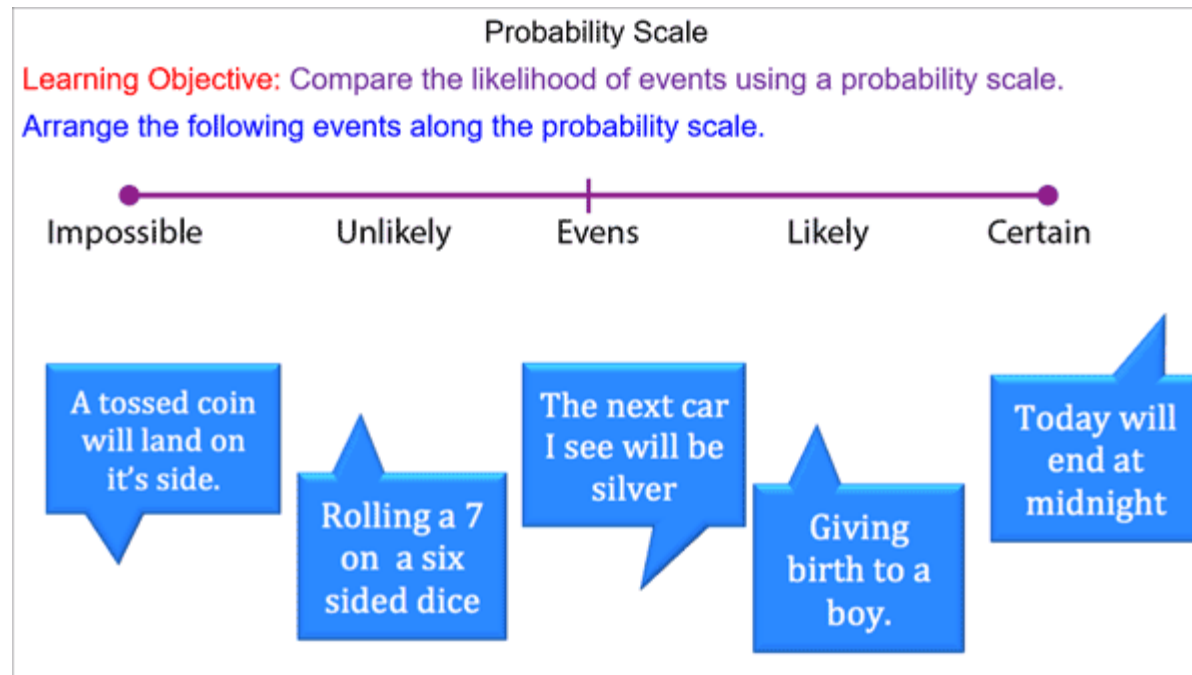


WHAT IS PROBABILITY

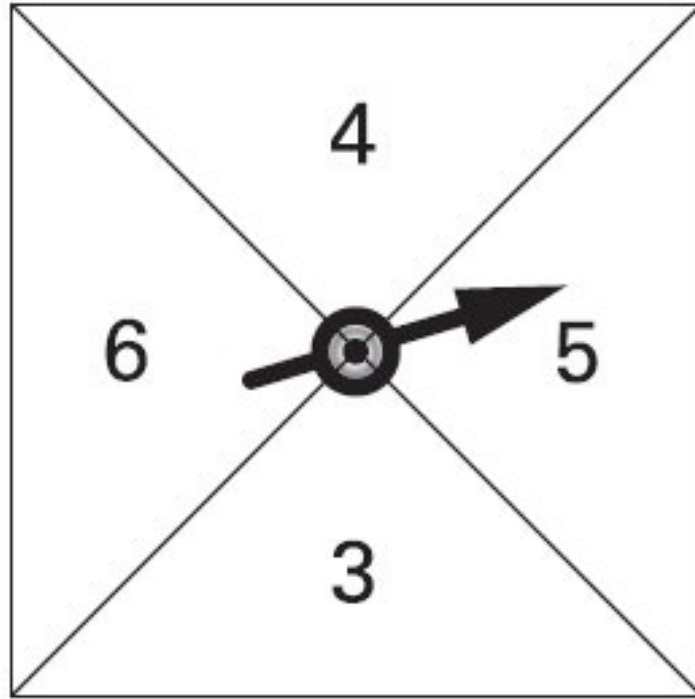
Probability or chance is **how likely** something is to happen:

If something has a **low** probability, it is **unlikely** to happen.

If something has a **high** probability, it is **likely** to happen.



Geeta has this spinner.



What is her chance of spinning the numbers in the boxes below?
Match each box to the correct word.

One has been done for you.



an odd number

2

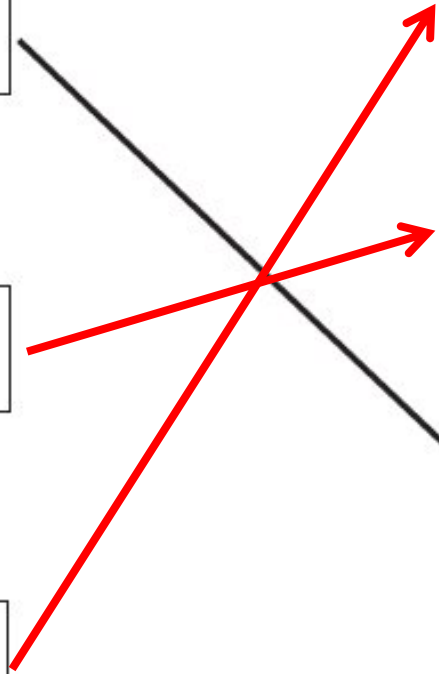
a number
less than 10

certain

impossible

even chance

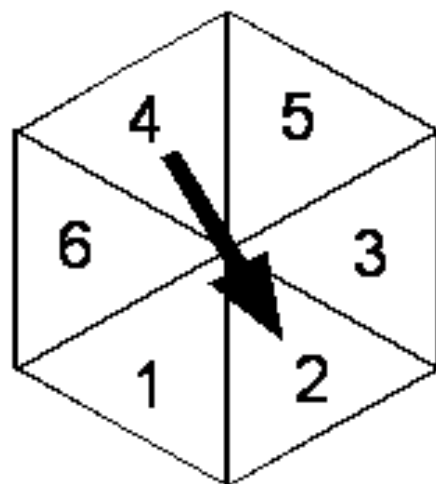
likely



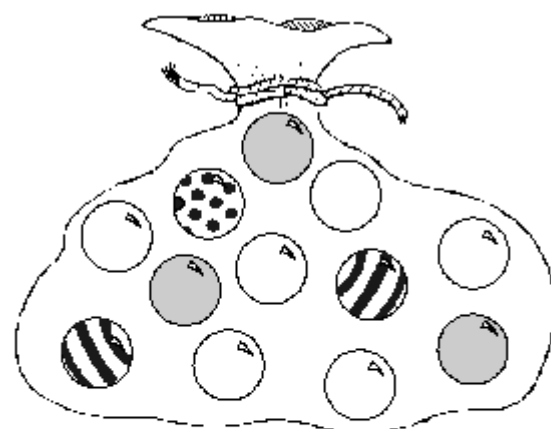
THE LANGUAGE OF PROBABILITY



You can spin any number from 1 to 6 on this spinner.



Write a number that is **impossible** to spin.



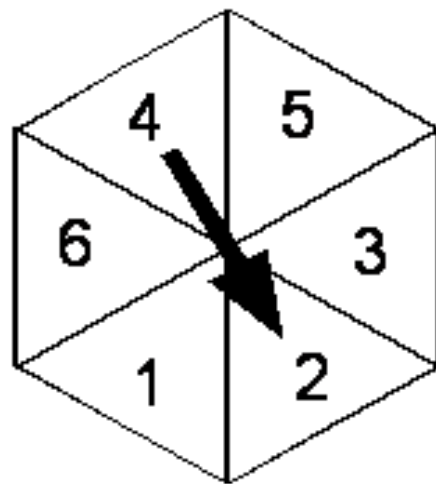
Key	
	striped
	spotty
	white
	grey

These marbles are hidden in a bag. The bag is shaken.

Pete pulls out one marble without looking.

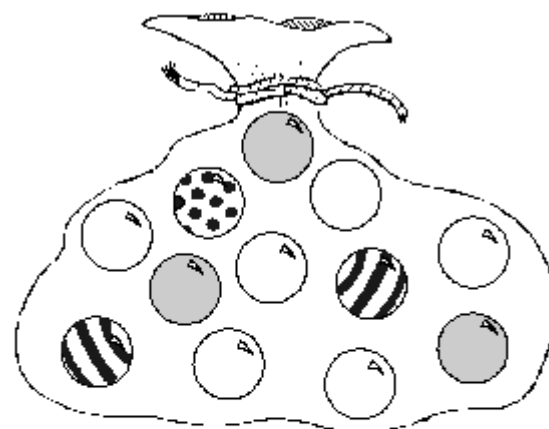
- (a) Which kind of marble is Pete most likely to pull out?
- (b) Explain how you know.

You can spin any number from 1 to 6 on this spinner.



Write a number that is **impossible** to spin.

Any number less than 1 or greater than 6.



Key	
	striped
	spotty
	white
	grey

These marbles are hidden in a bag. The bag is shaken.

Pete pulls out one marble without looking.

- (a) Which kind of marble is Pete most likely to pull out?
- (b) Explain how you know.

White, because there are more white marbles than any other colour.

Sample Space & Events

- The **sample space** (S) of a random phenomenon is a set of all possible outcomes of the random phenomenon.
- An **event** is a subset of the sample space.

Example: Flip a coin 3 times and record the side facing up each time.

- Sample space
 $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
- Events:
 - $\{\text{all heads}\} = \{HHH\}$
 - $\{\text{get one heads}\} = \{HTT, THT, TTH\}$
 - $\{\text{get at least two heads}\} = \{HHT, HTH, THH, HHH\}$

Intersections, Unions, and Complements

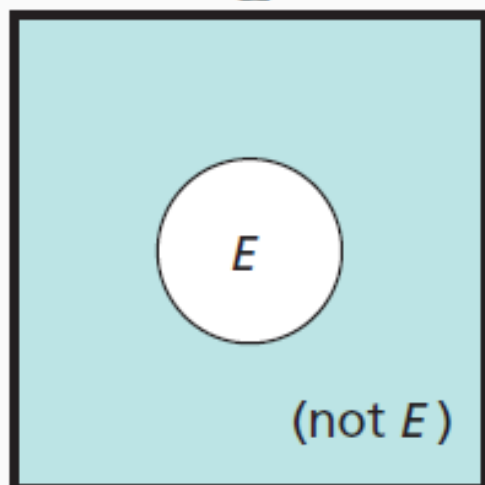
- We are often interested in events that are derived from other simpler events:
 - Rolling a 2 or 3
 - Patient who receives a therapy is relieved of symptoms and suffers from no side effects
- The event that **A does not occur** is called the **complement** of A and is denoted A^c or (not A)
- The event that **both A and B occur** is called the **intersection** and is denoted $A \cap B$ or (A and B)
- The event that **either A or B occurs** is called the **union** and is denoted $A \cup B$ or (A or B)

Venn Diagrams

Complements, intersections, unions of events can be represented visually using *Venn diagrams*:

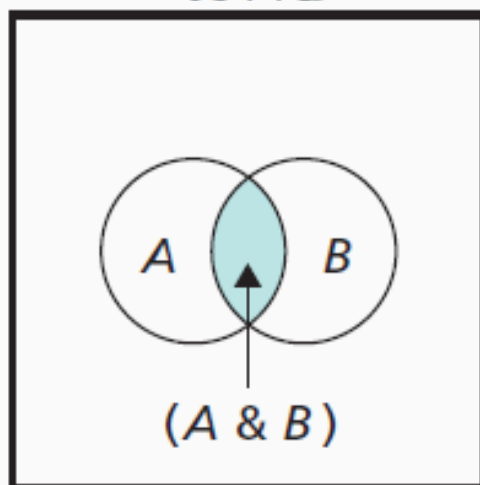
Complement

$$E^c$$



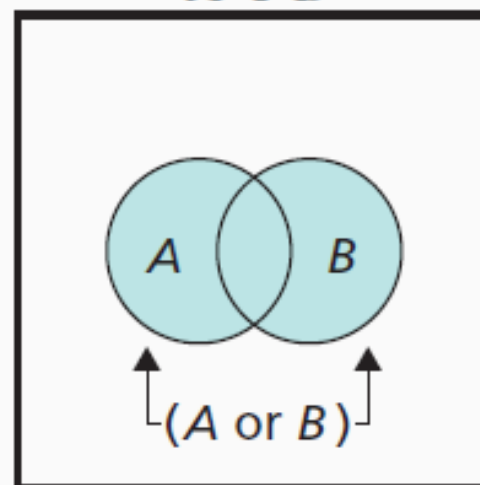
Intersection

$$A \cap B$$



Union

$$A \cup B$$



General Addition Rule

Disjoint Events (= Mutually Exclusive Events)

Disjoint (mutually exclusive) events cannot be both true.

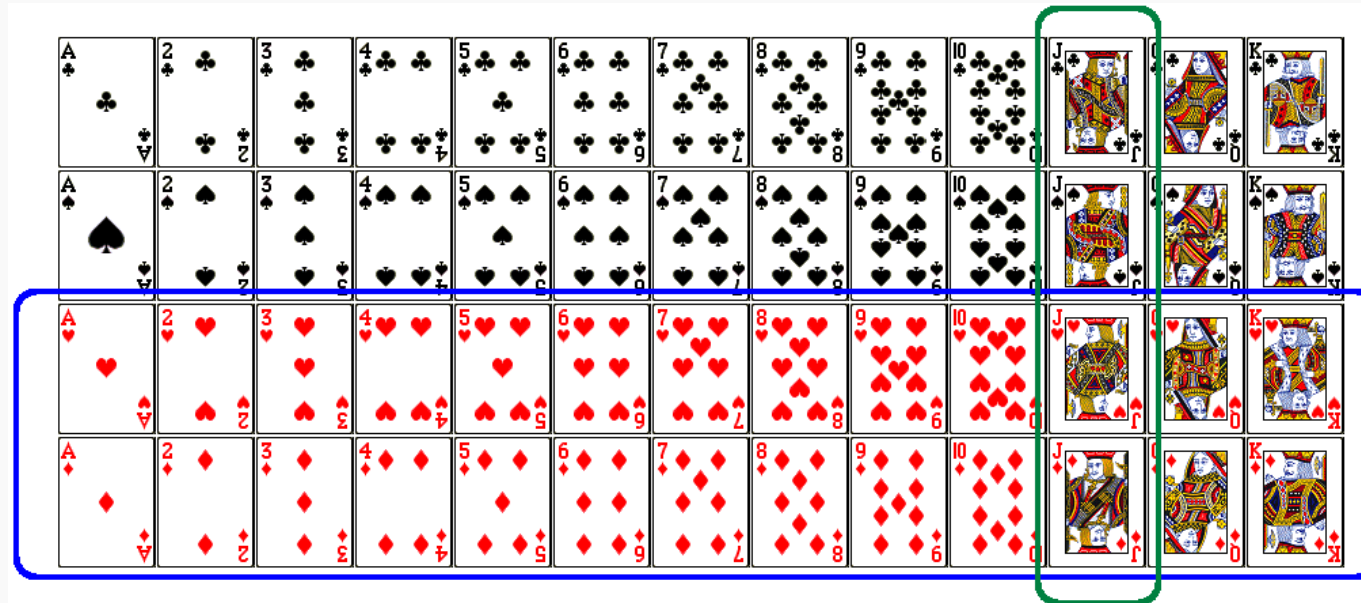
- Tossing a coin once, the events {getting H} and {getting T} are disjoint
- The events {Faheem passed SMMDS 5003} and {Faheem failed SMMDS 5003} are disjoint
- Drawing a card from a deck, the events {getting an ace} and {getting a queen} are disjoint

Non-disjoint events can be both true.

- The events {Faheem got an A in SMMDS 5003} and {Faheem got an A in DSTT 5001} are NOT disjoint

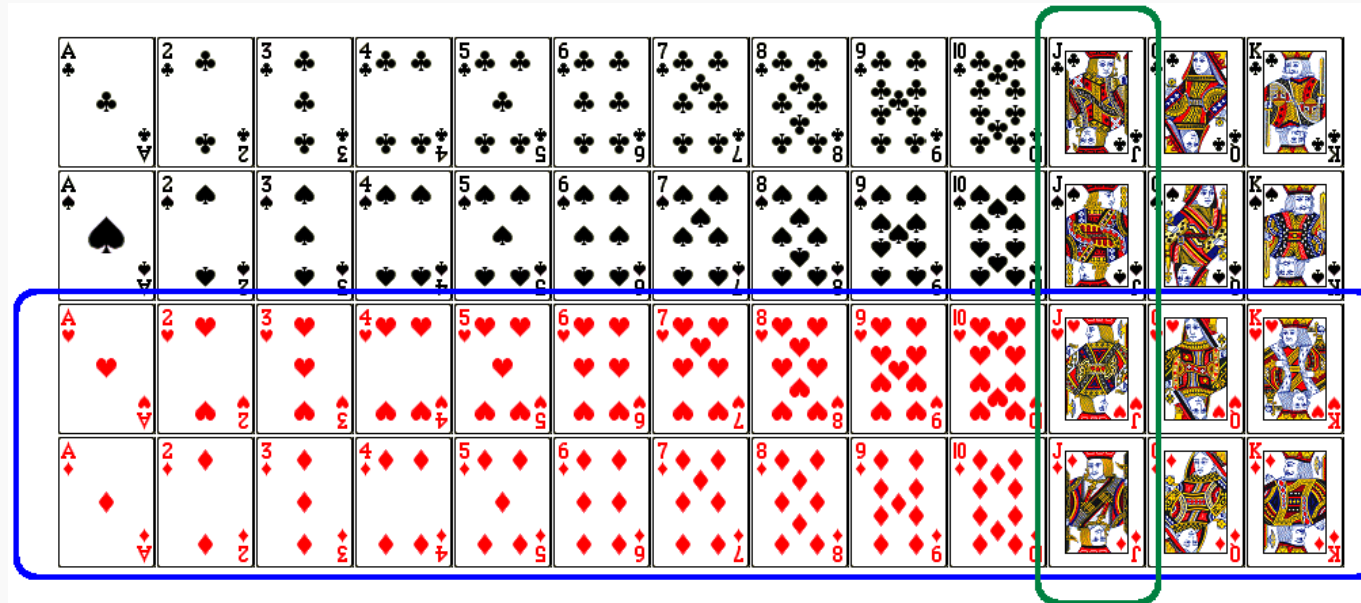
General Addition Rule

What is the probability of drawing a jack or a red card from a well shuffled full deck?



General Addition Rule

What is the probability of drawing a jack or a red card from a well shuffled full deck?



$$\begin{aligned} P(\text{jack or red}) &= P(\text{jack}) + P(\text{red}) - P(\text{jack and red}) \\ &= \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} \end{aligned}$$

General Addition Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

For disjoint events $P(A \text{ and } B) = 0$, so the above formula simplifies to

$$P(A \text{ or } B) = P(A) + P(B).$$

The Complement Rule

The Complement Rule

- Because an event must either occur or not occur,

$$P(A) + P(A^c) = 1$$

- Thus, if we know the probability of an event, we can always determine the probability of its complement:

$$P(A^c) = 1 - P(A)$$

- This simple but useful rule is called the **complement rule**

Example — The Complement Rule

Question: What is the probability of getting at least one head in 3 tosses of a fair coin?

- Sample space

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$\text{Event } A = \{\text{at least one heads}\}$$

$$= \{HHH, HHT, HTH, THH, HTT, THT, TTH\}$$

- $A^c = \{\text{not least one heads}\}$

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- $P(A^c) =$

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Event $A = \{\text{at least one heads}\}$

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- $A^c = \{\text{not least one heads}\} = \{\text{all tails}\} = \{TTT\}$
- $P(A^c) = 1/8$
- $P(A) = 1 - P(A^c) = 1 - 1/8 = 7/8$

Conditional Probability

Conditional Probabilities

Example – Poker

A card is drawn from a well-shuffled deck.

- What is the probability that the card drawn is a King?

$$P(\text{got a King}) = \frac{4}{52} = \frac{1}{13}.$$

- If the card drawn is known to be a face card (J, Q, K), what is the probability that it is a K?

Conditional Probabilities

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$$\frac{4}{12} = \frac{1}{3}$$

Conditional Probabilities

Given two events A and B. We denote the probability of event A happens **given** that event B is known to happen as

$$P(A \mid B),$$

read as the probability of “A given B.”

For the example on the previous slide, let

A = the card is a King,

B = the card is a face card (J,Q,K).

We have

$$P(A|B) = \frac{4}{12} \neq P(A) = \frac{4}{52}.$$

Current knowledge (face card) has changed (restricted) the sample space (possible outcomes).

Exercise

In the previous example, what is

$$P(B \mid A^c) = P(\text{the card is a face card} \mid \text{the card is not a King})?$$

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- If the card is not a King, as there are 4 Kings in a deck, then the card must be one of the remaining 48, among which 8 are face cards (4 Jacks and 4 Queens). So the answer is 8/48.

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- $= 1$, since a King is a face card.

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$$P(B \mid A) = P(\text{the card is a face card} \mid \text{the card is a King})?$$

- $= 1$, since a King is a face card.

$$P(A \mid B^c) = P(\text{the card is a King} \mid \text{the card is not a face card})?$$

- $= 0$, since a King is a face card. If it's not a face card, it cannot be a King.

Another Example

A deck of cards is well-shuffled and the two cards are drawn w/o replacement. What is the probability that second card is a King given that the first card is a King?

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given that the first card is NOT a King?

- $4/51$

Example: Age and Rank of Faculty Members

The table below cross-classifies faculty members in a certain university by age and rank.

Age (year)	Full professor	Associate professor	Assistant professor	Lecturer	Total
Under 40	54	173	220	23	470
40-49	156	125	61	6	348
50-59	145	68	36	4	253
60+	75	15	3	0	93
Total	430	381	320	33	1164

Marginal Probabilities of Contingency Tables

If a faculty member is selected at random, what is the probability that he/she is a full professor?

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$$P(\text{full professor}) = \frac{430}{1164} \approx 0.37.$$

Probabilities that involve only one of the categorical variables in a contingency table are called **marginal probabilities**.

Joint Probabilities of Contingency Tables

If a faculty member is selected at random, what is the probability that he/she is a full professor and under 40?

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- Probabilities that involve combination of categories of both categorical variables in a contingency table are called **joint probabilities**.
- same as the **overall proportions** for contingency tables

Conditional Probabilities of Contingency Tables

If we know that the selected faculty member is under 40, what is the probability that he/she is a full professor?

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Notice that the **conditional probabilities** of rank given age (column variable given row variable) for a randomly selected faculty member are simply the **row proportions** for contingency tables

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Notice that the **conditional probabilities** of age given rank (row variable given column variable) for a randomly selected faculty member are simply the **column proportions** for contingency tables

Conditional Probability

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The conditional probability of the outcome of interest A given condition B can be calculated as

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

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$$\begin{aligned} & P(\text{full prof. \& under 40}) \\ &= \frac{54/1164}{470/1164} \\ &= \frac{P(\text{full prof. \& under 40})}{P(\text{under 40})} \end{aligned}$$

If only the **joint probabilities** (relative frequencies) are available, but the counts are not,

Age (year)	Full prof.	Assoc. prof.	Assist. prof.	Lect.	Total
Under 40	0.046	0.149	0.189	0.020	
40-49	0.134	0.107	0.052	0.005	
50-59	0.125	0.058	0.031	0.003	
60+	0.064	0.013	0.003	0.000	
Total					

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Total	0.369	0.327	0.275	0.028	1.000

- the **marginal probabilities** can be computed from the joint probabilities

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- the **marginal probabilities** can be computed from the joint probabilities
- the **conditional probabilities** can be computed from the joint probabilities, e.g.,

$$P(\text{full prof.} \mid \text{under 40}) = \frac{P(\text{full prof. \& under 40})}{P(\text{under 40})} = \frac{0.046}{0.404}$$

General Multiplication Rule

General Multiplication Rule

The formula for conditional probability

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)}$$

can be used the other way around. Multiplied both sides by $P(A)$, we get the **General Multiplication Rule**:

$$P(A \text{ and } B) = P(A) \times P(B \mid A)$$

If we want $P(A \text{ and } B)$, and both $P(A)$, $P(B|A)$ are known or are easy to compute, we can use the General Multiplication Rule.

Example: General Multiplication Rule

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the probability that both cards are Kings?

Solution. Let

$A =$ 1st card is a King,

$B =$ 2nd card is a King.

- $P(A) = P(\text{the 1st card is a King}) =$.
- Given that the 1st card is a King, the conditional probability that the 2nd card is a King =? .
- So the probability that both cards are Kings =?

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Solution. Let

$A =$ 1st card is a King,

$B =$ 2nd card is a King.

- $P(A) = P(\text{the 1st card is a King}) = 4/52.$
- Given that the 1st card is a King, the conditional probability that the 2nd card is a King =? .
- So the probability that both cards are Kings =?

Example: General Multiplication Rule

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the probability that both cards are Kings?

Solution. Let

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- Given that the 1st card is a King, the conditional probability that the 2nd card is a King =? $P(B \mid A) = \frac{3}{51}.$
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$$P(A \text{ and } B) = P(A) \times P(B \mid A) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221} \approx 0.0045.$$

Example: General Multiplication Rule

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the probability that neither card is a K?

Sol. Let

$A =$ 1st card is NOT a K,

$B =$ 2nd card is NOT a K.

- $P(A) = P(\text{the 1st card is not a K}) =$.
- Given that the 1st card is not a K, the conditional probability that the 2nd card is not a K =? .
- So the probability that neither card is a K =?
- $P(\text{at least one of the two cards is a K}) =?$

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Sol. Let

$A =$ 1st card is NOT a K,

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- $$P(A \text{ and } B) = P(A) \times P(B \mid A) = \frac{48}{52} \times \frac{47}{51} = \frac{188}{221} \approx 0.851.$$
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- $P(\text{at least one of the two cards is a K}) = ?$

$$\{\text{at least a K}\}^c = \{\text{neither is a K}\}$$

$$\text{So, } P(\text{at least a K}) = 1 - P(\text{neither is K}) = 1 - 0.851 = 0.149.$$

General Multiplication Rule for Several Events

$$P(ABC) = P(A) \times P(B \mid A) \times P(C \mid AB)$$

$$P(ABCD) = P(A) \times P(B \mid A) \times P(C \mid AB) \times P(D \mid ABC)$$

$$P(ABCDE) = P(A) \times P(B \mid A) \times P(C \mid AB) \times P(D \mid ABC) \times P(E \mid ABCD)$$

and so on

Example: General Multiplication Rule for Several Events

Five cards are dealt from a deck of well-shuffled card. What is the chance that none of them are hearts ♥?

Sol. Let A_i be the event that the i th card dealt is not a ♥.

- $P(A_1) = P(\text{1st card is not a ♥}) = 39/52$

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- Given that the 1st card is not a ♥, the conditional probability that the 2nd is not a ♥ = $P(A_2 \mid A_1) = \frac{38}{51}$.

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- Given neither of the first two cards is a ♥, the conditional probability that the 3rd is not a ♥ = $P(A_3 \mid A_1A_2) = \frac{37}{50}$.
- Likewise, $P(A_4 \mid A_1A_2A_3) = \frac{36}{49}$ $P(A_5 \mid A_1A_2A_3A_4) = \frac{35}{48}$

By the General Multiplication Rule,

$$P(A_1A_2A_3A_4A_5) = \frac{39}{52} \times \frac{38}{51} \times \frac{37}{50} \times \frac{36}{49} \times \frac{35}{48} \approx 0.222$$

Continue the previous slide, what is the probability of getting at least one heart ♥ among the five cards?

Sol. Since $\{\text{at least one ♥}\}^c = \{\text{no ♥}\}$, by the complement rule,

$$P(\text{at least one ♥}) = 1 - P(\text{no ♥}) = 1 - 0.222 = 0.778.$$

Keep in mind:

- The complement of $\{\text{at least one is...}\}$ is $\{\text{none is...}\}$
- The complement of $\{\text{all are...}\}$ is $\{\text{at least one is not ...}\}$

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- Knowing that the coin landed on a head on the first toss does not provide any useful information for determining what the coin will land on in the second toss.
 - Outcomes of two tosses of a coin are independent.
- Knowing that the first card drawn from a deck is an ace does provide useful information for determining the probability of drawing an ace in the second draw.
 - Outcomes of two draws from a deck of cards (w/o replacement) are dependent.

Independence and Conditional Probabilities

In mathematical notation, if $P(A \mid B) = P(A)$ then the events A and B are said to be independent.

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In mathematical notation, if $P(A|B) = P(A)$ then the events A and B are said to be independent.

- Conceptually: Giving B doesn't tell us anything about A .

Equivalently, one can also check the independency of the events A and B by check whether $P(B \mid A) = P(B)$.

Practice – Checking for Independence

Between January 9-12, 2013, SurveyUSA interviewed a random sample of 500 NC residents asking them whether they think widespread gun ownership protects law abiding citizens from crime, or makes society more dangerous. 58% of all respondents said it protects citizens. 67% of White respondents, 28% of Black respondents, and 64% of Hispanic respondents shared this view. Are opinion on gun ownership and race ethnicity independent?

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$P(\text{protects citizens})$ varies by race/ethnicity, therefore opinion on gun ownership and race ethnicity are dependent.

<http://www.surveyusa.com/client/PollReport.aspx?g=a5f460ef-bba9-484b-8579-1101ea26421b>

Multiplication Rule for Independent Events

When A and B are independent

$$P(A \text{ and } B) = P(A) \times P(B)$$

- This is simply the general multiplication rule:

$P(A \text{ and } B) = P(A) \times P(B \mid A)$ in which $P(B \mid A)$ reduce to $P(B)$
when A and B are independent

- More generally,

$$P(A_1 \text{ and } \dots \text{ and } A_k) = P(A_1) \times \dots \times P(A_k)$$

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Exercise: You roll a 6-face die twice, what is the probability of getting two aces in a row?

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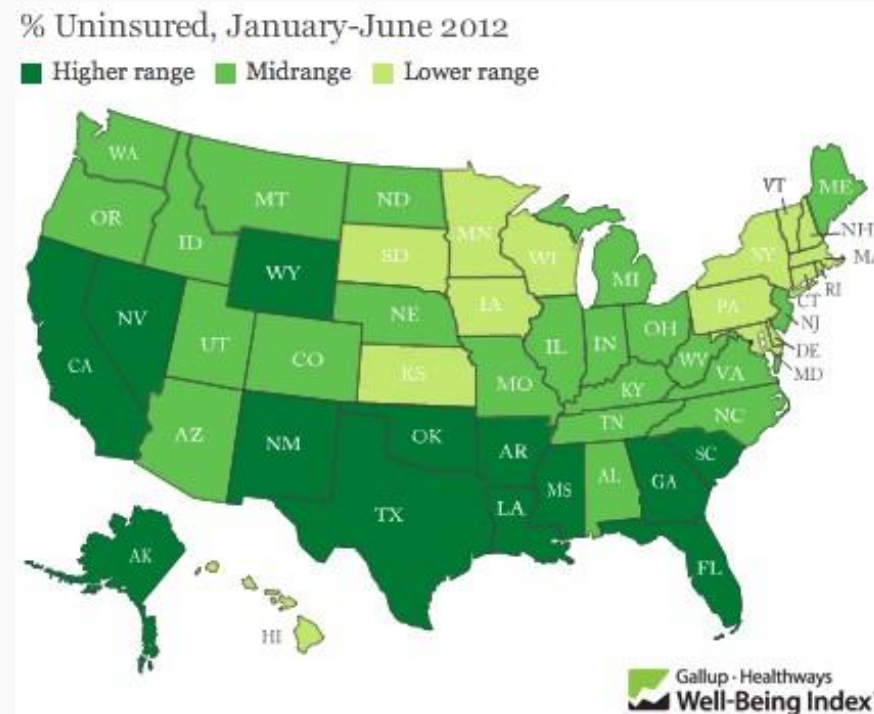
Exercise: You roll a 6-face die twice, what is the probability of getting two aces in a row?

$$P(\text{ace in the 1st roll}) \times P(\text{ace on the 2nd roll}) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Practice

A recent Gallup poll suggests that 25.5% of Texans do not have health insurance as of June 2012. Assuming that the uninsured rate stayed constant, what is the probability that two randomly selected Texans are both uninsured?

- (a) 25.5^2
- (b) 0.255^2
- (c) 0.255×2
- (d) $(1 - 0.255)^2$

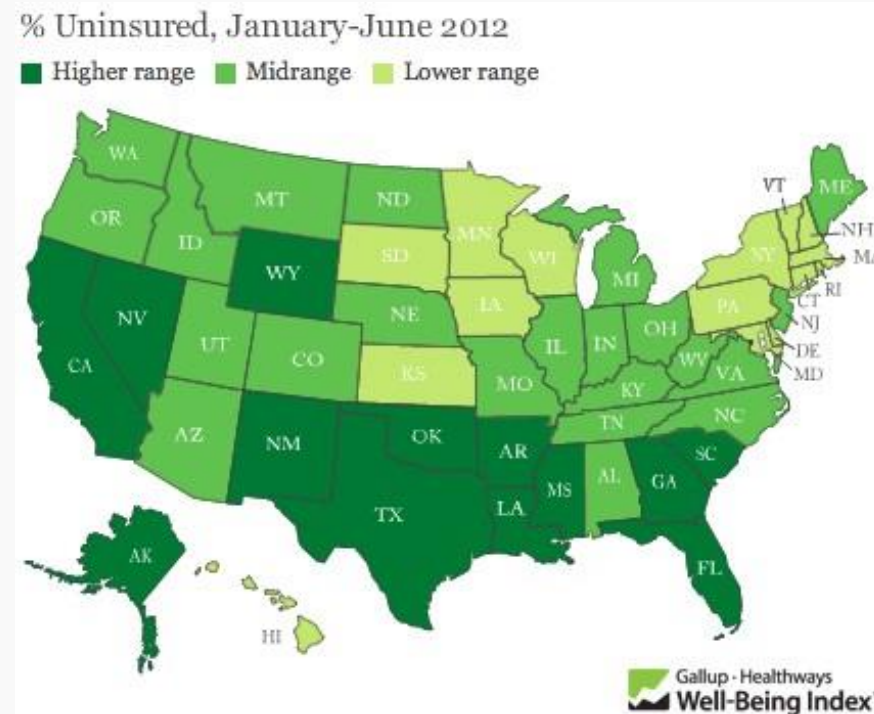


[http:// www.gallup.com/ poll/ 156851/ uninsured-rate-stable-across-states-far-2012.aspx](http://www.gallup.com/poll/156851/uninsured-rate-stable-across-states-far-2012.aspx)

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Putting everything together...

If we were to randomly select 5 Texans, what is the probability that at least one is uninsured?

By the complement rule:

$$\begin{aligned}P(\text{at least 1 uninsured}) &= 1 - P(\text{none uninsured}) \\&= 1 - [(1 - 0.255)^5] \\&= 1 - 0.745^5 \\&= 0.77\end{aligned}$$

Keep in mind that

$$\begin{aligned}P(\text{at least one}) &= 1 - \\&P(\text{none})\end{aligned}$$

Abuse of the Multiplication Rule

As estimated in 2012, of the U.S. population,

- 13.4% were 65 or older, and
- 52% were male.

True or False and explain: $0.134 \times 0.52 \approx 7\%$ of the U.S. population were males age 65 or older.

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True or False and explain: $0.134 \times 0.52 \approx 7\%$ of the U.S. population were males age 65 or older.

False, age and gender are dependent.

In particular, as women on average live longer than men, there are more old women than old men.

Among those age 65 or older, only 44% are male, not 52%.

Of the U.S. population in 2012, only $0.134 \times 0.44 \approx 5.9\%$ were males age 65 or older.

Tree diagrams and Bayes' Theorem

Example – Nervous Job Applicant

Suppose an applicant for a job has been invited for an interview.

The probability that

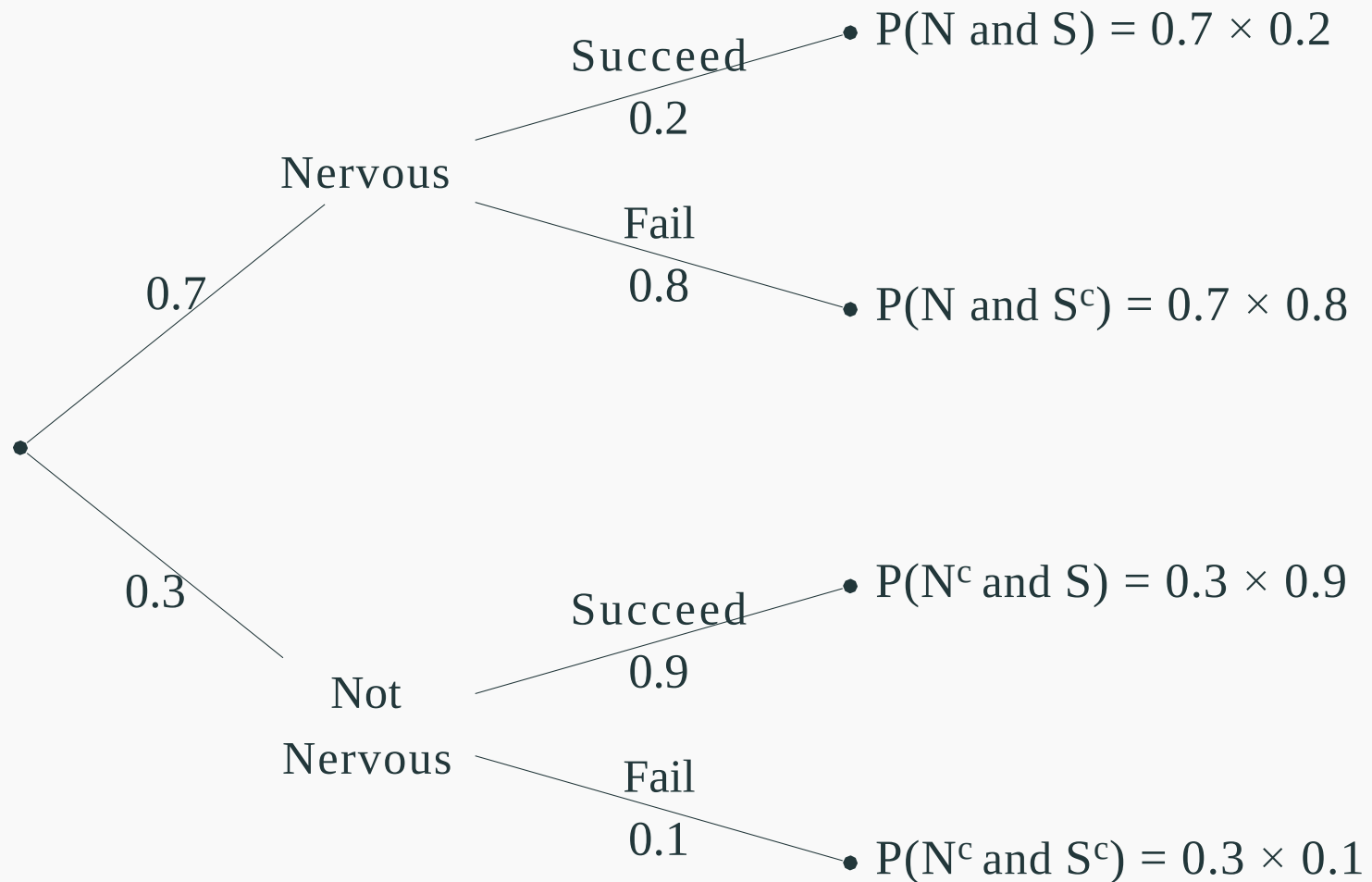
- he is nervous is $P(N) = 0.7$,
- the interview is successful given he is nervous is $P(S|N) = 0.2$,
- the interview is successful given he is not nervous is $P(S|N^c) = 0.9$.

What is the probability that the interview is successful?

$$\begin{aligned} P(S) &= P(S \text{ and } N) + P(S \text{ and } N^c) \\ &= P(N) P(S | N) + P(N^c) P(S | N^c) \\ &= 0.7 \times 0.2 + 0.3 \times 0.9 = 0.41 \end{aligned}$$

Tree Diagram for the Nervous Job Applicant Example

Another look at the nervous job applicant example:



Nervous Job Applicant Example Continued

Conversely, given the interview is successful, what is the probability that the job applicant is nervous during the interview?

$$\begin{aligned} P(N \mid S) &= \frac{P(N \text{ and } S)}{P(S)} = \frac{0.41}{0.41} \\ &= \frac{P(N) P(S \mid N)}{0.41} \\ &= \frac{0.7 \times 0.2}{0.41} = \frac{14}{41} \approx 0.34. \end{aligned}$$

in which $P(S) = 0.41$ was found in the previous slide.

Bayes' Theorem

The problem in the previous slide is an example of the **Bayes' Theorem**.

Knowing $P(B \mid A)$, $P(B \mid A^c)$, and $P(A)$, is there a way to know $P(A \mid B)$?

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

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$$\begin{aligned} P(A | B) &= \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A)P(B | A)}{P(B)} \\ &= \frac{P(A)P(B | A)}{P(B \text{ and } A) + P(B \text{ and } A^c)} \end{aligned}$$

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Medical Testing

A common application of Bayes' rule is in the area of diagnostic testing

- Let D denote the event that an individual has the disease that we are testing for
- Let $T+$ denote the event that the test is **positive**, and $T-$ denote the event that the test comes back **negative**
- $P(T+ | D)$ is called the **sensitivity** of the test
- $P(T- | D^c)$ is called the **specificity** of the test
- Ideally, both $P(T+ | D)$ and $P(T- | D^c)$ would equal 1.

However, diagnostic tests are not perfect.

They may give false positives and false negatives.

Enzyme Immunoassay Test for HIV

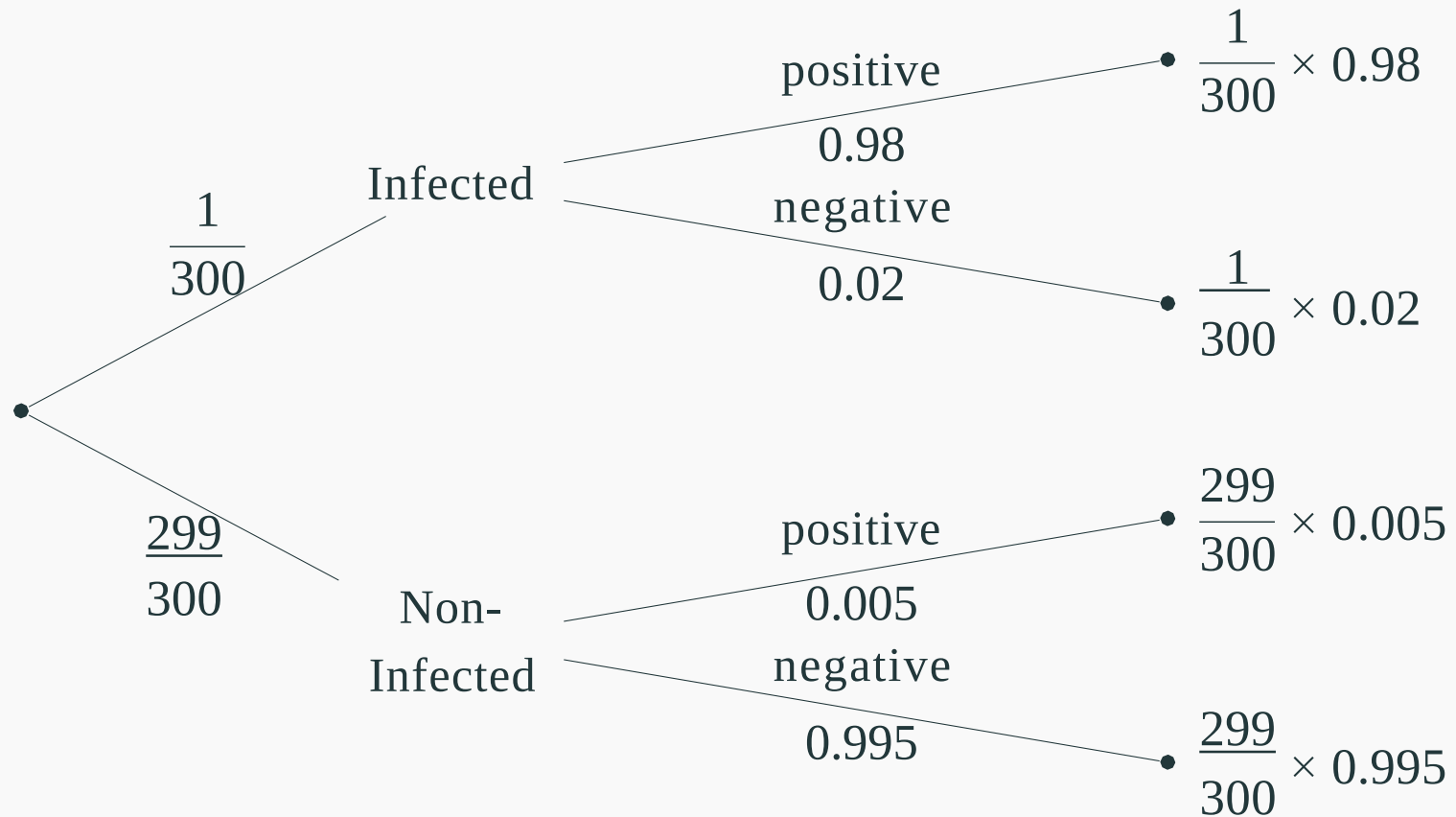
- $P(T + | D) = 0.98$ (sensitivity - positive for infected)
- $P(T - | D^c) = 0.995$ (specificity - negative for non-infected)
- $P(D) = 1/300$ (prevalence in the US: estimated 1 million HIV infected)

What is the probability that the tested person is infected if the test was positive?

$$\begin{aligned} P(D | T+) &= \frac{P(D)P(T+ | D)}{P(D)P(T+ | D) + P(D^c)P(T+ | D^c)} \\ &= \frac{(1/300) \times 0.98}{(1/300) \times 0.98 + (299/300) \times 0.005} \\ &= 39.4\% \end{aligned}$$

This test is not confirmatory. Need to confirm by a second test.

Tree Diagram for the Enzyme Immunoassay Test for HIV



$$P(D = T+) = \frac{(1/300) \times 0.98}{(1/300) \times 0.98 + (299/300) \times 0.005}$$

Draw a tree diagram!

Don't try to memorize the formula of Bayes' Theorem.